

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

BCH 201 — Past Questions & Materials
by Renance

Course: BCH 201
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 200

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Paper 1 of 2 — BCH 201 / EMT 212 / BCT 211

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
COLLEGE OF SCIENCE
DEPARTMENT OF ENVIRONMENTAL MANAGEMENT AND TOXICOLOGY
FIRST SEMESTER 2024/2025 UNDERGRADUATE EXAMINATION

BCH 201 / EMT 212 / BCT 211: Introductory Biochemistry (3 units)

Time: 2 hours

INSTRUCTION: Answer any four (4) questions

1.
 - (a) Water is an essential component of life, and its importance cannot be overstated. Explain this statement by outlining four (4) critical functions of water that contribute to its indispensable role.
 - (b) Calculate the pH and pOH of a solution with a hydrogen ion concentration, $[H^+]$, of 2.4×10^{-5} M.
2.
 - (a) Describe the general structure of an amino acid.
 - (b) Proteins exhibit structural organization at four hierarchical levels. Briefly discuss any two (2) of these structural levels.
3.
 - (a) Define carbohydrates.
 - (b) List the major types of carbohydrates.
 - (c) What are epimers? Provide examples with structural representations to illustrate your answer.
4.
 - (a) Define nucleic acids.
 - (b) Illustrate the basic structures of purine and pyrimidine bases.
 - (c) What are nucleotides?
5.

Biological membranes serve purposes beyond merely acting as physical barriers. Discuss this statement by highlighting the diverse roles performed by membranes in living organisms.
6.

Differentiate between the unit membrane hypothesis and the fluid mosaic model, comparing them from both structural and functional perspectives.

Paper 2 of 2 — BCH 201 I EMT I 212 I BCT I 211 INTRODUCTORY BIOCHEMISTRY

Level 200 • First Semester • PastQuestion

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